

Paraneoplastic Dermatoses

There is an evolving consensus regarding what constitutes a true paraneoplastic syndrome. Two essential criteria must be met:

1. The dermatosis must develop after the onset of the malignant tumor, even if the tumor is asymptomatic or occult.
2. The dermatosis and the malignancy must follow a parallel course—complete removal of the tumor leads to resolution of the skin condition, and tumor recurrence results in relapse of the dermatosis.

A classification of clinicopathologic categories is outlined in Table 154-1.

Epidemiology

Skin changes can serve as markers of internal malignancy, although the exact prevalence is unknown due to variability in disease presentation and the rarity of certain conditions. The risk of developing a paraneoplastic dermatosis correlates with age, sex, and established risk factors for the associated internal malignancy.

The strength of the association between a specific dermatosis and cancer varies. For example, **hypertrichosis lanuginosa acquisita** is linked to a broad range of malignancies, whereas **necrolytic migratory erythema (NME)** is highly specific for glucagon-producing pancreatic neuroendocrine tumors. Specific malignancy associations for each paraneoplastic dermatosis are detailed in Table 154-1.

Etiology and Pathogenesis

Paraneoplastic dermatoses are classified by their primary clinical features. A more precise pathophysiologic classification is limited by incomplete understanding of underlying mechanisms.

Establishing a definitive link between a cutaneous eruption and an internal malignancy is easier when both the skin condition and the tumor are rare. Challenges arise when the dermatosis is common—such as **seborrheic keratoses** in **Leser-Trélat sign**—and is associated with a wide range of frequently occurring cancers, leading to numerous anecdotal reports.

Hyperkeratotic Dermatoses

Acanthosis Nigricans (See Chap. 152)

The term *acanthosis nigricans* was introduced by Unna, though the first cases linked to malignancy were described independently by Pollitzer and Janovsky in 1890. Curth classified acanthosis nigricans into malignant, benign, syndromic, and pseudoacanthosis nigricans (associated with obesity).

Epidemiology

Approximately 80% of acanthosis nigricans cases are idiopathic or associated with benign conditions such as endocrinopathies or hereditary disorders. Malignancy-associated acanthosis nigricans is rare, occurring in 1 of 35 patients with intrathoracic or intra-abdominal malignancies and in 2 of 12,000 cancer patients overall. It predominantly affects individuals over 40 years of age, with at least 80% of malignant cases occurring in this group.

Paraneoplastic Dermatoses: At a Glance

Skin manifestations can signal internal malignancy through:

- Epi-phenomenon
- Familial cancer syndromes
- Direct tumor extension

Key Features of Paraneoplastic Dermatoses:

- Follow a parallel course with the underlying malignancy
- Enable early diagnosis and disease monitoring

Dermatoses strongly associated with internal malignancy: - Tripe palms

- Bazex syndrome
- Paraneoplastic pemphigus
- Erythema gyratum repens
- Necrolytic migratory erythema
- Hypertrichosis lanuginosa acquisita

Dermatoses weakly associated with internal malignancy: - Acanthosis nigricans

- Acquired ichthyosis
- Multiple seborrheic keratoses

Management: Treatment of the dermatosis centers on identifying and managing the underlying malignancy.